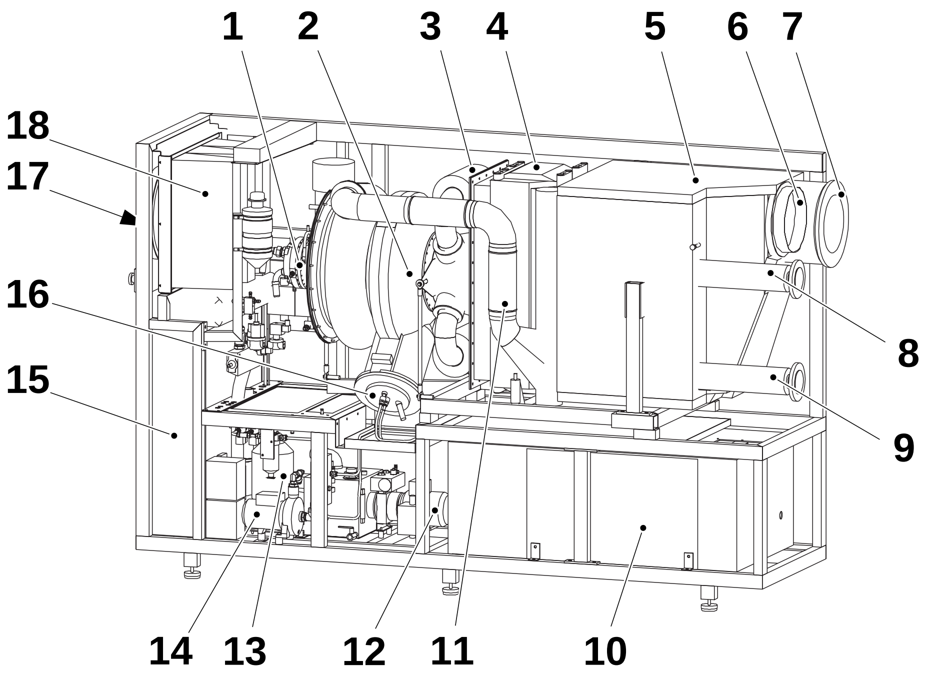
T100 microturbine CHP system

Technical description

1. Product description



1. Generator
2. Gas turbine engine
3. Pipe from recuperator
4. Recuperator
5. Exhaust gas heat exchanger
6. Ventilation air outlet
7. Exhaust outlet
8. Water inlet
9. Hot water outlet
10. Power electronics
11. Pipe to recuperator
12. Oil pump
13. Buffer air pump
14. Cooling water pump
15. Control system
16. Combustion chamber
17. Air inlet
18. Air filter

1.1 Main components

Turbec T100 microturbine CHP system is a Combined Heat and Power unit. The unit produces electricity and heat fuelled by natural gas. The system is designed for indoor installation and takes air from an outside intake. The CHP unit is divided into following main parts:

• Gas turbine engine
• Electrical generator
• Electrical system
• Exhaust gas heat exchanger
• Supervision and control system

Gas turbine engine

The gas turbine is a single shaft engine. The main components are:

• Housing
• Compressor
• Recuperator
• Combustion chamber
• Turbine

Housing

The electrical generator and the rotating components of the gas turbine are mounted on the same shaft. The two engine parts together with the shaft are mounted in the same housing.

Compressor

The Turbec T100 uses a radial centrifugal compressor to compress ambient air. The pressure ratio is about 4.5:1. The compressor is mounted on the same shaft as the turbine and the electrical generator.

Recuperator

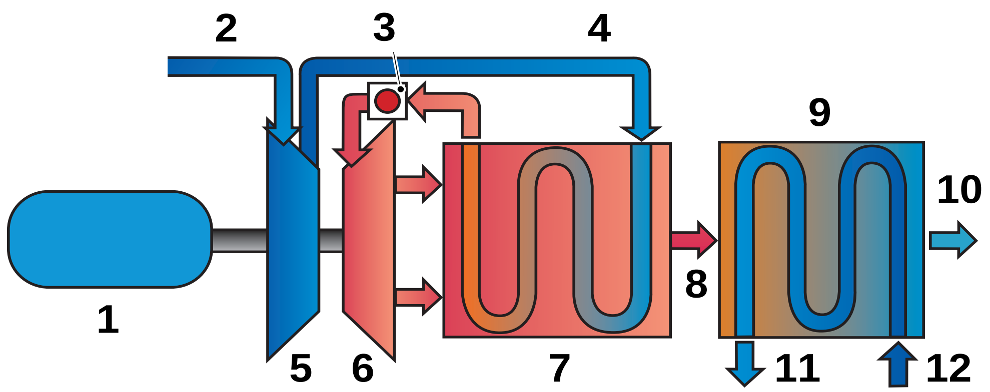
The recuperator is a gas to air heat exchanger attached to the microturbine. The heat is exchanged from the hot exhaust gases to the compressed air that is fed to the combustion chamber. The recuperator increases the electrical efficiency of the gas turbine.

Combustion chamber

The preheated compressed air is mixed with the natural gas. During start up an electrical igniter in the combustion chamber ignites the mixture. The combustion chamber is of a lean pre-mix emission type, achieving low emissions of NOx, CO and unburned hydrocarbons in the exhaust gases.

Turbine

The turbine drives the compressor and the generator. When the combustion gases leave the combustion chamber the temperature is approx. 950°C (1742°F) and pressure is approx. 4.5 bar (65 psi). As the combustion gases expand through the turbine the pressure decreases to close to atmospheric pressure and the heat drops to approx. 650°C (1202°F).



1. Generator
2. Inlet air
3. Combustion chamber
4. Air to Recuperator
5. Compressor
6. Turbine
7. Recuperator
8. Exhaust gases
9. Exhaust gas heat exchanger
10. Exhaust gas outlet
11. Hot water outlet
12. Water inlet

Electrical generator

The electric power is generated with the rotating permanent magnet. The rotor is suspended by two bearings on each side of the permanent magnet. The generator acts as an electric starter for the gas turbine to bring the CHP unit into operation. The generator is designed for high conversion efficiency and is water-cooled.

Electrical system

Before the generated AC reaches the grid it needs to be rectified and transformed to the preferred frequency. The AC is first rectified to DC and then converted to a three-phase AC. An inductor stabilizes the AC output. An EMC filter protects the operation to prevent generated interference. The electrical system is entirely controlled and automatically operated by the Power Module Controller (PMC).The electrical system is used in reverse when it works as the electric starter to the gas turbine.

Exhaust gas heat exchanger

The exhaust gas heat exchanger is of gas-water counter-current flow type. The temperature of the exhaust gases is approx. 270°C (518°F), entering the exhaust gas exchanger. The thermal energy from the exhaust gases is transferred to the hot-water system by the exhaust gas heat exchanger. The outlet water temperature depends on the incoming water conditions, temperature and mass flow.

The exhaust gases leave the exhaust gas heat exchanger through an exhaust pipe and the subsequent chimney.

Supervision and control system

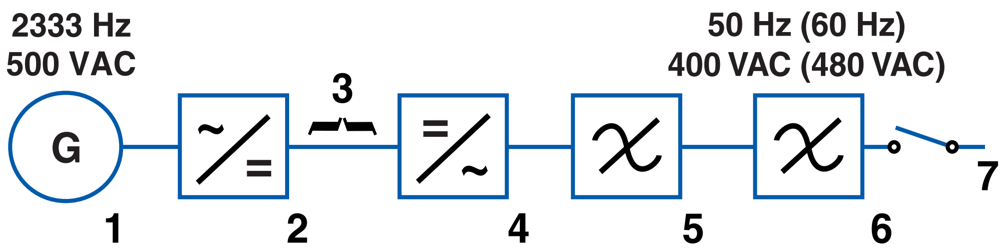
The Turbec T100 is controlled and supervised with an automatic control system. The CHP unit needs no attendance of personnel in normal use. In case of critical distortion the system automatically shuts down and records the fault to the PMC. The PMC is used to start, stop and supervise the CHP unit. The PMC runs the system by using values from sensors monitoring:

- Heat demand
- Gas pressure
- Oil temperature
- Vibrations

If a critical fault should appear in the CHP unit the PMC is programmed to execute one of following actions:

- Normal stop
- Emergency stop

The fault is presented on the display on the control panel and logged in the system.



1. Generator
2. Rectifier/ Start converter
3. DC bus
4. Converter
5. Line filter
6. EMC filter
7. Main circuit breaker

1.2 Auxiliary systems

- Lubrication oil system
- Cooling system
- Air intake and ventilation system
- Fuel gas system
- Buffer air system
- Fuel gas compressor
- Remote control

Lubrication oil system

The purpose of the system is to lubricate the squeezed film bearings of the unit. The system consists of a closed piping system and a lubrication oil tank placed within the enclosure. A motor-driven pump circulates the lubrication oil. The heated lubrication oil circulates from the bearings to an oil-to-air cooler and is cooled to 50°C (122°F). The PMC system monitors oil pressure and oil temperature in filters, oil preheater (before start-up) and lubrication oil tank to safeguard a reliable and continuous operation of the microturbine unit.

Cooling system

The generator and the electrical system are cooled with a separate cooling water system placed within the enclosure.

Air intake and ventilation system

The CHP unit draws ambient air from an outdoor intake. As the air reaches the CHP unit the airflow is divided into two partial flows. The main flow acts as combustion air in the microturbine. The other flow ventilates the excess heat out from the CHP unit. An induced draft fan placed on the outside wall creates a negative pressure inside the Turbec T100 enclosure, for safety reasons and for cooling.

The ventilation air leaves the CHP unit through an air duct. Due to the relatively large diameter of the air duct the T100 can be placed several meters from the outside wall. There are two air filters in the Turbec T100 system, the coarse prefilter (optional supply) is placed close to the outdoor intake of the air and the fine filter is placed close to compressor within the enclosure for filtering the combustion air. The full air requirement is approx. 1.7 kg/s, (3.7 lb/s) distributed on approx. 0.9 kg/s (1.9 lb/s) for ventilation and approx. 0.8 kg/s (1.8 lb/s) for combustion.

Fuel gas system

The fuel gas system includes piping from the gas entrance at enclosure, auto shut off valve, filter, fuel block, pressure sensor, fuel control valves and pipes to injectors. The inlet gas system is designed with fuel gas evacuation. The fuel system fulfils applicable gas installation- and safety standards.

Buffer air system

The buffer air system includes an air pump, interconnecting piping and filter for removal of oil mist. The air pump circulates compressed air to the sealing system to block the lubrication oil from entering the gas turbine engine. After the sealings, the air passes through an oil filter in which the oil mist is separated from the air and collected in the lubrication oil tank.

Fuel gas compressor

If the natural gas pressure is below 6 bar (g) (87 psig), it is necessary to use a fuel gas compressor to raise the pressure. The fuel gas compressor is situated outside the CHP unit.

Remote control

The control system is fully automated but can be remote controlled via the modem or with a PC network. The remote control allows an operator to remotely start and stop, download supervisory data and set values on the CHP unit to the same extent as a local operator can do on the operator's panel.

Additional emergency stop

The T100 is prepared to meet the requirements of an extra wall mounted emergency stop.

1.3 Auxiliary options

BMS

An already site existing BMS (Building Management System) can act as a complementary to the remote control system submitted by Turbec. This feature gives means of adjusting the electrical power output, obtaining a run and fault signal, starting and stoping the CHP unit via a BMS.

External power metering

The T100 is prepared for external power metering to measure the delivered power net to the grid. The nominal power value can be displayed on the operator's panel and via the remote control.

1.4 Technical data

General identification

Type:	Microturbine
Manufacturer:	Turbec AB, Sweden
Model:	T100
Application:	Combined Heat and Power
Usage:	Indoors
Dimensions of CHP unit:	Width 840 mm (33.1") Height 1 920 mm (75.6") Length 2 900 mm (114.2")
Weight:	2000 kg (4410 lb)
Fuel:	Natural gas
Ambient inlet, air temperature:	–25°C to 40°C (–13°F to 104°F)
Ambient inlet humidity:	100%
Surrounding air temperature:	±0°C to 40°C (32°F to 104°F)
Surrounding humidity:	80%

Gas turbine

Compressor type:	Centrifugal
Turbine type:	Radial
Type of combustion chamber:	Lean pre-mix
Number of combustion chambers:	1
Pressure in combustion chamber:	4.5 bar (a) (65 psia)
Turbine inlet temperature:	950°C (1742°F)
Number of shaft:	1
Nominal speed:	70 000 rpm
Consumption of lubrication oil:	<9 litre/6 000 h operation (<2.4 gal/6 000 h operation)

Electrical data

Frequency output:	50 Hz (60 Hz)
Max. allowed mains frequency variation:	± 5%
Max. allowed mains voltage variation:	± 10%
Adjustable power factor:	0.80 leading to 0.80 lagging
Nominal voltage output:	400/230 VAC, 3 phases (480/277 VAC)
Start up voltage:	400 VAC, 50 Hz (480 VAC, 60 Hz)
Start up power:	Max 15 kW
Harmonic current:	
Maximum total distortion:	5 %
Maximum single distortion:	3 %
Output circuit:	5 wire connection
Protection circuit containing:	Over/under frequency trip* Over/under voltage trip* Phase error* Short circuit current protection Ground fault detection *(*Located in power electronics)*

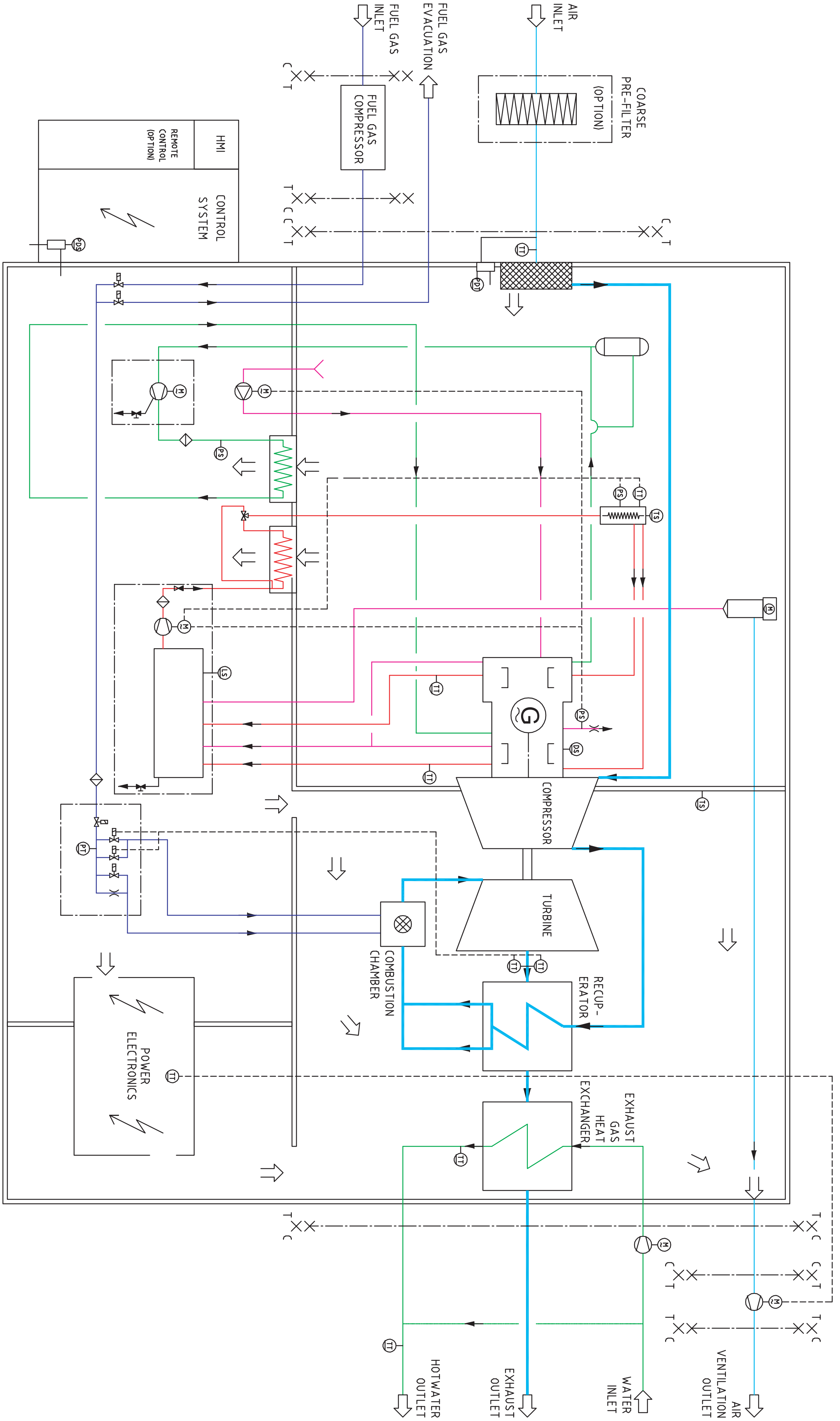
Gas requirements

Pressure min/max:	6/8.5 bar (g) (87/123 psig)
Temperature min/max:	0°C/60°C (32°F/140°F)
Wobbe index:	43-55 MJ/m_n^3 (1154-1476 Btu/scf)
Fuel gas flow:	Depending on gas composition
Example at nominal load, 100 kW:	
Fuel gas LHV:	*39 MJ/m_n^3 (1047 Btu/scf)*
Volume flow:	*31 m_n^3/h (1095 scf/h)*

Gas compressor

Compressor type:	Scroll compressor
Inlet gas pressure:	0.02 - 1.0 bar (g) (0.3 - 14.5 psi)
AC Power supply:	345 - 525 VAC, (50/60 Hz)
Noise level	75 dBA at 1 m (3.3 ft)
Dimensions:	Width 610 mm (24") Height 1 070 mm (42") Length 1 370 mm (54")

1.5 Principal diagram



2. Scope of Supply & Terminal Points

2.1 Scope of Supply

The delivery covers a complete Combined Heat and Power unit, ready for installation at terminal points below. The main parts of the scope of supply are as follows:

- Microturbine unit including turbine, compressor, generator and recuperator
- Fuel gas system
- Fuel gas compressor, delivered as a separate item
- Closed cooling water system for generator and electrical system
- Hot water temperature sensor
- Closed cooling system for gas turbine
- Closed lubrication system for gas turbine
- Exhaust gas heat exchanger for hot water production
- Enclosure for insulation of heat and noise
- Induced draft fan for ventilation of the enclosure, delivered as a separate item
- Frequency and voltage converters for the supply of 400 VAC, 3 phase, 50 Hz (480 VAC, 3 phase, 60 Hz)
- High performance electrical output filtering
- Automatic voltage regulation
- Circuit breaker
- Starting system including synchronizer
- Control panel with digital LCD display
- Computer modem for remote operation.
- Documentation:
  - Operation- and maintenance manual
  - Installation manual

2.2 Options

- Coarse pre filter
- Additional relay protection (if required)

2.3 Terminal points

The following terminal points applies to the scope of delivery of this proposal:

- Connections for in- and outlet for the water on the exhaust gas heat exchanger, located at the T100 enclosure wall.
- Connection for air inlet, located at the T100 enclosure wall.
- Coarse pre filter (allowed to be bought from other supplier)
- Connection to exhaust gas outlet located at the T100 enclosure wall.
- Fuel gas evacuation system outlet
- Connections to fuel gas compressor with terminal points on the fuel gas compressor enclosure wall.
- Main inlet to gas fuel system, located at the T100 enclosure wall.
- Outlet connection of the ventilation air system, located at the T100 enclosure wall.
- Electrical connection at the T100 enclosure wall and connection to ventilation fan, hot water pump, hot water temperature, fuel gas compressor start signal, network cable and phone line for modem connection.

2.4 Terminal Points (option)

- Extra connection at the T100 enclosure wall for BMS, external emergency stop and external power meter.

3. Performance

Performance notes

The performance data includes all the T100 auxiliary consumption, i.e. cooling water pump, oil pump, induced draft fan, fuel gas compressor, buffer air pump, and are based on the following conditions for new and clean equipment operating at ISO standard conditions:

Site elevation:	0 m above sea level
Ambient temperature:	15°C (59°F)
Relative humidity:	60%
Pressure drop to air inlet flange:	0 Pa (0 psi)
Pressure drop from exhaust gas flange:	0 Pa (0 psi)

3.1 Performance data

Net electrical output:	100 kW*
Net electrical efficiency:	30 %*
Net total efficiency:	80 %
Fuel consumption:	333 kW (1 136 000 Btu/h)
Net thermal output (hot water):	167 kW (570 000 Btu/h)
Exhaust gas flow:	0.79 kg/s (6370 lb/h)
Exhaust gas temperature:	55°C (131°F)
Water inlet temperature:	50°C (122°F)
Water outlet temperature:	70°C (158°F)
Noise level:	70 dBA at 1 meter (3.3 ft)
Fuel type:	Natural gas
Fuel gas pressure:	0.02 - 1.0 bar (g) (0.3 - 14.5 psi)

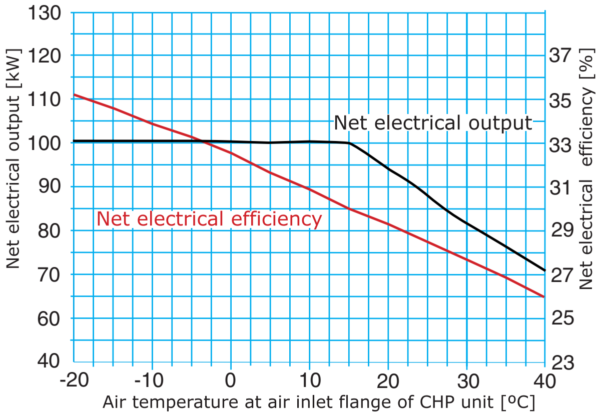
Volumetric exhaust gas emissions at 15% O_2:

	100% load
NO_x:	< 15 ppm v
CO:	< 15 ppm v
UHC:	< 10 ppm v

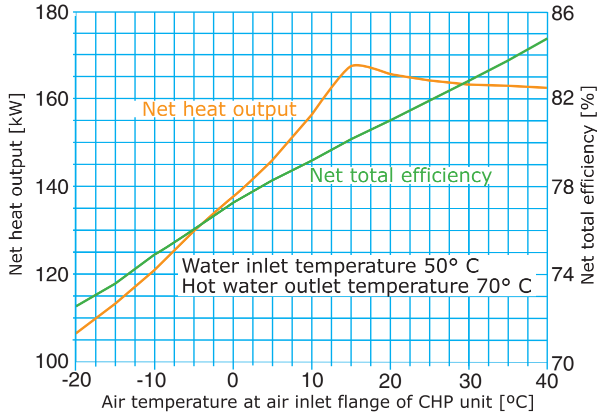
(Guaranteed data at LHV = 39 000 kJ/m_n (1047 Btu/scf), *Measurement tolerance to be considered)*

3.2 Air inlet temperature influence

Electrical output and efficiency



Heat output and total efficiency

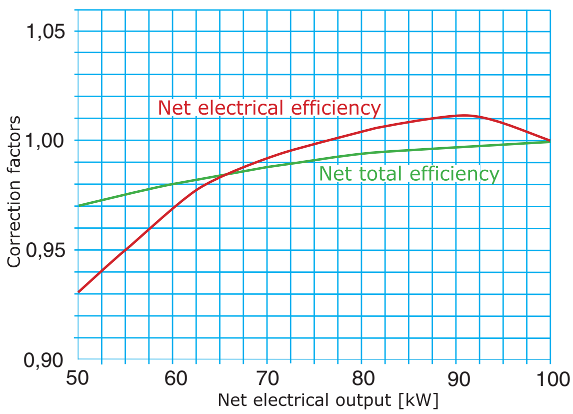


Corrections of thermal output at different water temperatures

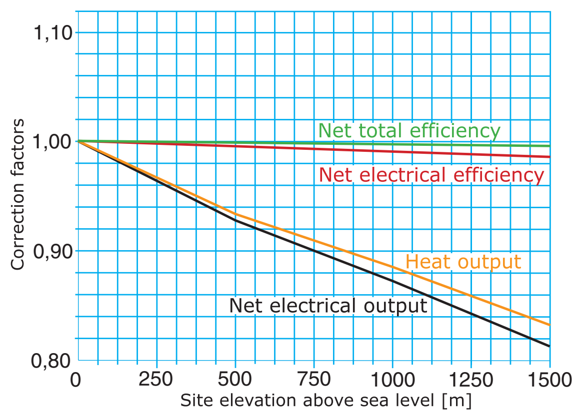
The thermal output statement in the diagram above depends on the inlet water temperature and on the difference between outlet and inlet water. A decrease (or increase) of 10°C (18°F) of the inlet water temperature will increase (or decrease) the thermal output by approximately 8 kW.

3.3 Corrections of performance

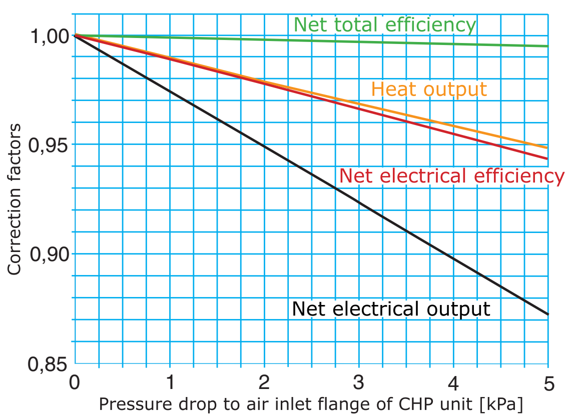
Part load performance



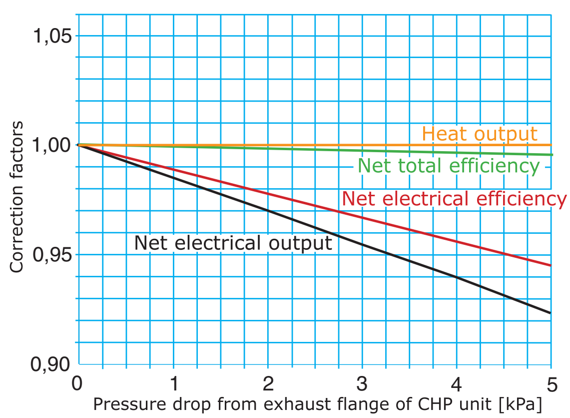
Performance at different site elevations



Performance at different inlet pressure drop

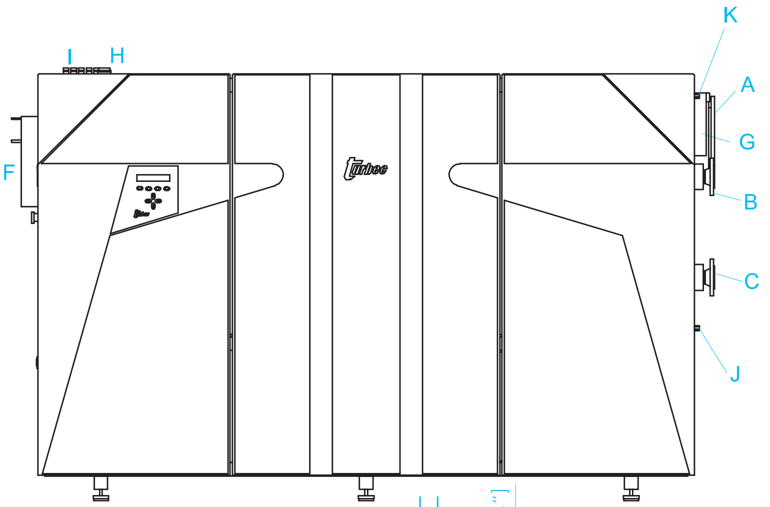


Performance at different exhaust pressure drop

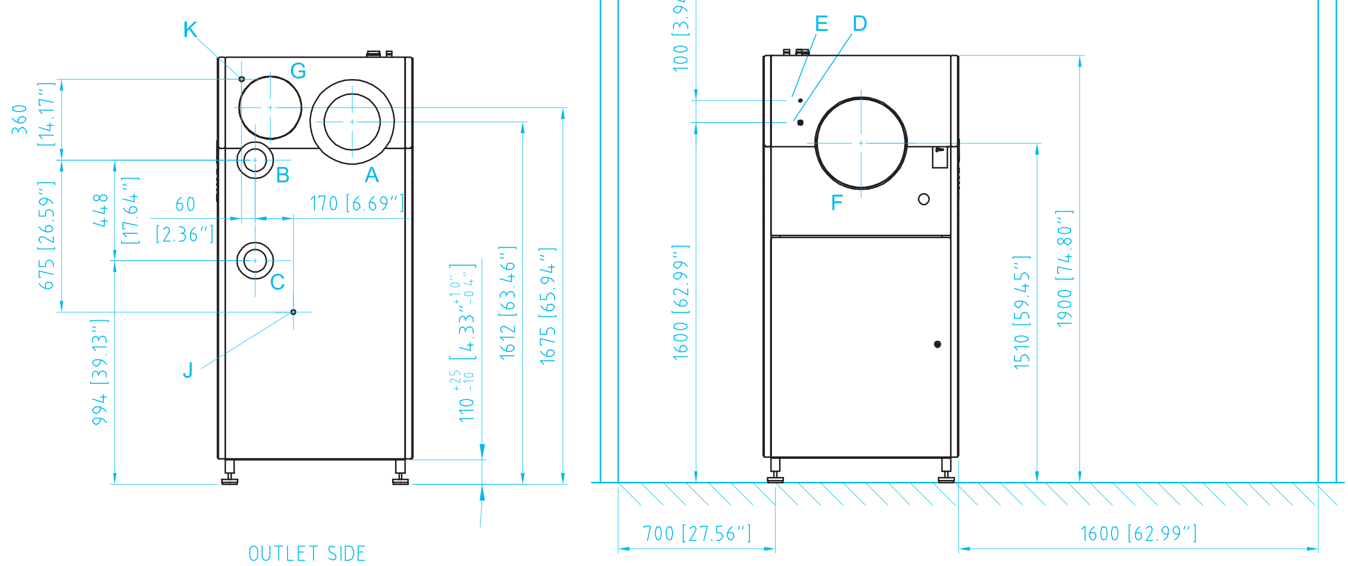


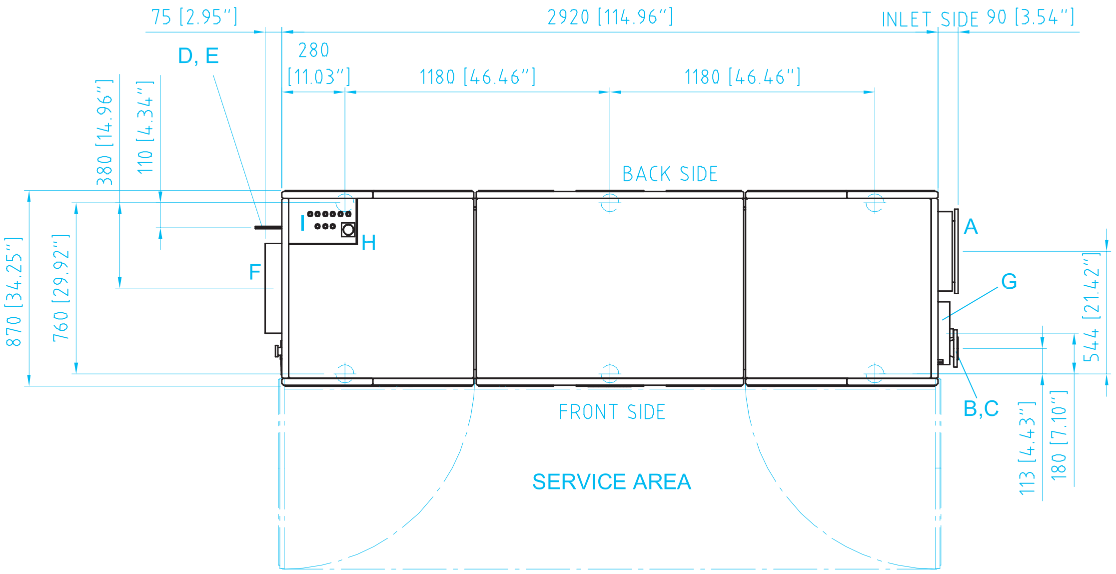
4. Installation

4.1 Layout & terminal points



A.	Exhaust gas outlet
B.	Water inlet
C.	Hot water outlet
D.	Fuel gas inlet
E.	Evacuation fuel gas outlet
F.	Pre-filtered air inlet
G.	Ventilation outlet
H.	Grid cable
I.	Electrical cables
J.	Heat exchanger, drain
K.	Heat exchanger, air evacuation





Note!

This is an installation guideline. Complete installation information, is provided in the installation manual.

4.2 Installation guidelines

4.2.1 Combustion and ventilation air inlet pipe installation

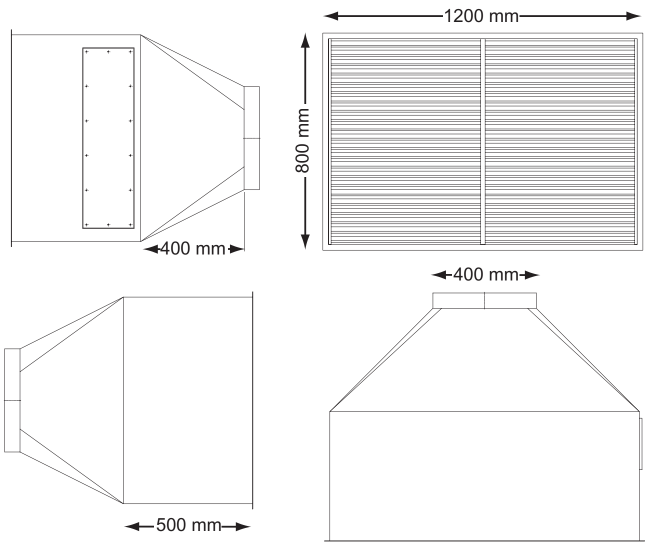
Dimension of connection:	Ø 400 mm (15.75")
Connection type:	Twisted tube galvanized
Max air flow at 15°C (59°F):	1,69 kg/s (=1,38 m^3/s) 3.73 lb/s (48.7 ft^3/s)
Max air pressure drop from pipe inlet to CHP unit*:	< 180 Pa (0.72" W.C.)
Max air pressure drop from pipe inlet to CHP unit with worn filter*:	< 320 Pa (1.29" W.C.)

(The example corresponds to 10 m pipe, Ø 400 mm (15.75") and two 90° elbows)*

Installation on site:
Pipe, Ø 400 mm (15.75"), to be installed from ambient air intake to the interface of T100 CHP unit. If possible mount through a northern wall.
Pre filter mounted in ambient air intake, size about 1200 x 800 mm (3.94 x 2.62 ft).
The pre filter is available as an option from Turbec or can be supplied from local supplier if the requirements needed are observed:

Filter type:	G3, (EU3)
Min air flow at 15°C (59°F):	1.69 kg/s (=1.38 m^3/s) 3.73 lb/s (=48.7 ft^3/s)
Max pressure drop for a clean filter:	< 50 Pa (0.20" W.C.)
Max pressure drop for a worn filter:	< 190 Pa (0.76" W.C.)

Make sure the filter type preferred uses an intake screen, E. g. use two standard size cases with the dimensions: 610 x 610 mm (2.00 x 2.00 ft).



The pipe shall be fitted with 50 mm (2") condense insulation.

4.2.2 Ventilation outlet installation

Dimension of connection:	Ø 280 mm (11.0")
Material of connection:	Twisted tube
Max air flow:	0,90 kg/s (=0,80 m^3/s) 1.98 lb/s (28.2 ft^3/s)

Installation on site:
The pipe, Ø 280 mm (11.0"), to be installed between T100 CHP unit to the ventilation system exit.

Induced draft fan mounted to an ambient air outlet. For safety reasons it can not be mounted on the CHP unit. (Fan included in Turbec scope of supply). The ventilation outlet has to be designed so that rain and snow cannot enter the CHP unit.

4.2.3 Exhaust gas outlet installation

Type of connection:	Flange, DN250, PN6 DIN 86044
alternative	*(10", ANSI B16.5, 300 lbs)*
Material of connection:	Black steel
Max exhaust gas flow:	0,87 kg/s (1.92 lb/s)
Max temperature:	325°C (612°F)

Installation on site:
Pipe (Ø 273 mm) to be installed from the T100 CHP unit to a chimney, material according to national standard.
Design the exhaust gas installation so the pressure drop is kept low. An expansion bellow has to be installed after the T100 CHP interface. The chimney outlet has to be designed so that rain and snow cannot enter the CHP unit.

4.2.4 Hot water installation

Type of connection:	Flange, DN50, PN40 Ø 60,3 mm
alternative	*(2", ANSI B16.5, 300 lbs)*
Min in water temperature:	50°C (122°F)
Max out water temperature:	150°C (302°F)
Max water pressure:	24 bar (g) (348 psig)

Example:

In water temperature:	50°C (122°F)
Out water temperature:	70°C (158°F)
Water flow:	2,1 l/s (4.4 ft^3/min)
Pressure loss for water in exhaust gas heat exchanger:	16 kPa (2.3 psi)

Installation on site:
The CHP hot water circuit is connected to the existing system with one inlet pipe and one outlet pipe. A water-circulating pump is to be installed in the pipe before the T100 CHP unit. The pump size is site specific with a default protective motor switch dimensioned for 0.37 kW/400 VAC. Isolation valves and an inlet strainer is mandatory.

4.2.5 Electrical installation

Connection block for:

- Grid cable
- Power to induced draft fan
- Power to water pump
- Start signal fuel gas compressor
- Temperature sensor in water outlet
- External emergency stop
- External power metering
- Local remote control

Suggested installation on site, local:

Grid cable: Cu 4 x 120 + 70mm² (AWG 4, AWG 2)
Max fuse 200A gG
Max short circuit current 35 kA
Min/max Ø 42/50mm

Main circuit breaker example: ABB OESA 250 D4PL

Cable from T100 CHP unit to:

Screened cable to
ventilation fan: Cu 4 x 1,5mm² (AWG 16)
Hot water pump: Cu 4 x 1,5mm² (AWG 16)
Hot water temperature: Cu 2 x 0,75mm² (AWG 18)
Fuel gas compressor
start signal: Cu 2 x 0,75mm² (AWG 18)
Phone line for modem connection: RJ 11
Optional:
External emergency stop: Cu 2 x 0,75mm² (AWG 18)
External power meter: Cu 2 x 0,75mm² (AWG 18)

The remote control system can be connected to a PC on site or via modem.

Note:
Some parts of T100 is designed to be live when the T100's internal circuit breaker is swiched off. For safety reasons, install an external maun circuit breaker as a visual and lock00able break point and to assure the T100 is dead, before any work on the CHP is started.

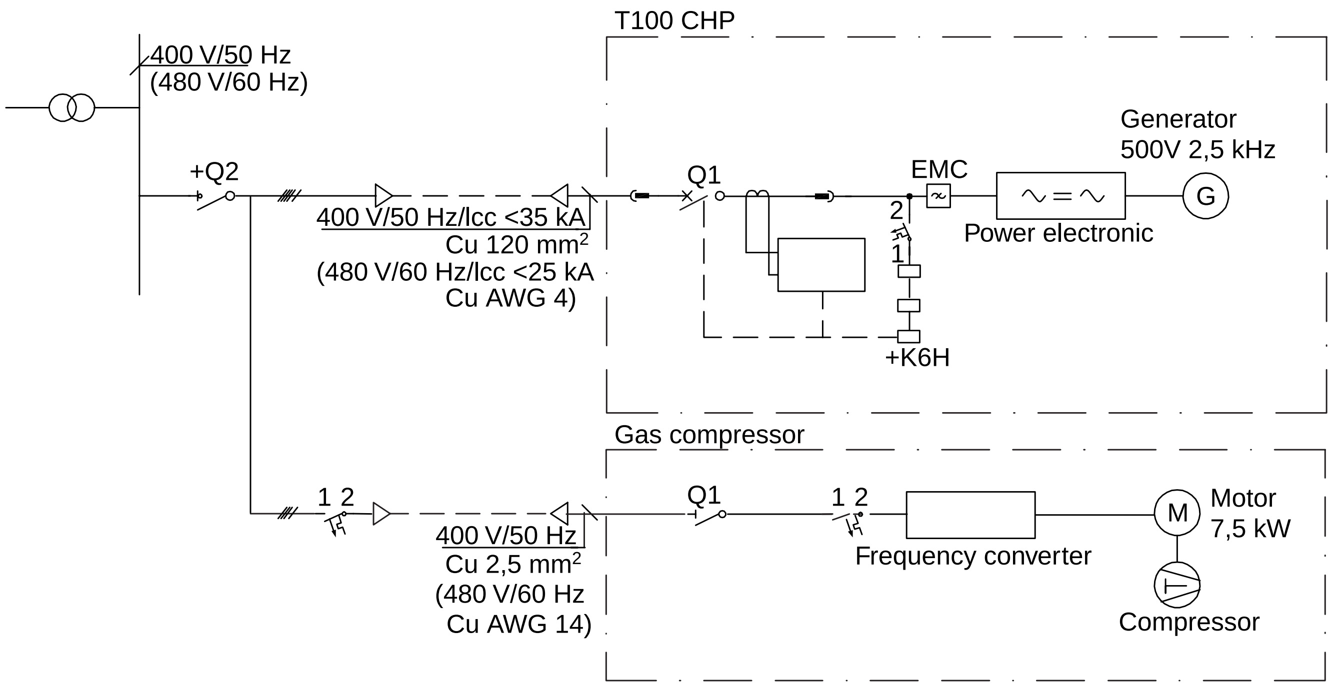
A skilled and experienced electrician may recommend an alternative installation.
The electrical system is subject to several site-specific demands. Those are both technical and regulatory, and have to be adapted to the standards and regulations in each country.

4.2.6 Natural gas installation

Type of connection: Flexible tube, Ø 12/10 mm
Natural gas pressure: 6 bar (g) to 8,5 bar (g)
(87 psig to 123 psig)
Fuel gas evacuation outlet: Ø 10/8 mm

Installation on site:
The natural gas piping is routed from the low-pressure system to the fuel gas compressor, from there it is routed to the CHP unit.
The CHP unit is equipped with the fuel gas evacuation, there has to be a 10-mm pipe routed from the unit to the outside of the building.

Note:
The complete fuel gas installation is subject to local regulations and may therefore need to be adapted to the national standards and regulations.



4.4 CE compliance

This equipment complies with the basic health and safety regulations of the European Economic Area (EEA). The T100 CHP unit follows these directives:

Machinery Directive 98/37/EC
Electromagnetic Compatiblity Directive 89/336/EEG
with amendments 92/31/EEG and 93/68/EEG
Low Voltage Directive 73/23/EEC, 93/68/EEC.

5. Maintenance & Operation

5.1 Maintenance concept

Lifetime of components

The simple and rugged design of the T100 power module provides for a durable operation during many years. Expected life times of main components are listed below:

Gas turbine engine	> 60 000 hrs
Recuperator:	> 60 000 hrs
Combustion chamber:	> 30 000 hrs
	(some parts < 30 000 hrs)

Maintenance concept

Inspections, maintenance and overhaul intervals and duration

	Interval (h)	Outage (h)
Inspection	6 000	24
Overhaul	30 000	48

Inspection and maintenance

- General checks
- Inspection of the combustion chamber and the change-out of the fuel nozzle
- Change of consumables, filters, oil and water refill (if necessary)
- General visual check
- Fuel gas compressor inspection and refill of oil and change-out of oil filter
- Check/change of 24 VDC battery

Overhaul

- Same procedure as in Inspection
- Change of the entire combustion chamber
- Engine refurbishment
- Change of bearings in lubrication oil pump, ventilation fan and buffer air pump

The overhaul will be performed in a change-out program, whereas the power module of the package, that is the generator, compressor and turbine section.

The replaced unit will be refurbished, which includes change-out of the entire combustion chamber and of other parts that need to be replaced due to corrective measures.

Preventive maintenance will be based on the continuous diagnostics of the Microturbine, performed remotely through the modem. Any necessary measure will then be done during the next Inspection, allowing as low as possible outage time. The overhaul program is an option within the contract, and it can be dropped if the service is performed by local service technichans.

5.2 Operation

Start

The starting system is fully automatic, engaged by a push button on the local control panel or via the remote control system. The start up sequence includes starting, ventilation, warm up, loading and synchronization to the grid.

Normal start-up procedure will bring the T100 Microturbine into 100% load in 240 seconds. The start-up sequence will use power from the grid with the generator running as motor until enough loading is reached to ignite the combustion chamber.
The enclosure and the engine and exhaust pipe changes air completely before ignition.

Restart time after shutdown will be less than 5 minutes, provided that the engine still is warm. The T100 is equipped with a function for automatic restart, activated when the stop was caused by external disturbances.

Operating Mode

The T100 microturbine operates on standard speed regulation, which means that the CHP unit maintains a high efficiency in part load operation.

The normal operating mode is that the heat production overrules power production. A sensor measures the temperature of the outgoing water or appropriate water on site, and the load of the T100 is adjusted in order to meet the set value given by the operator. In this mode, the power output will depend on the heat load.

Control System

The T100 CHP unit can be controlled directly from the operator's panel, or remotely via field bus or modem connection. Due to the high degree of automation, input values are kept to a minimum.

- Start and Stop of the machine, including a manual emergency stop.
- Control of power output and outlet water temperature.
- Fault reset.

The hot water system runs only when the CHP unit is active. The circulation pump begins when the CHP unit starts. The T100 CHP unit can be remote controlled via a modem and PC computer from any location or later via a local network.

Signals received from the CHP unit are of different categories.

- Operation mode: stopped, starting, in service, and stopping.
- Measuring data: electric power, (net electrical power as option), hot water temperature, produced electricity, and running hours, etc.
- Warnings.
- Faults with description.

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Most data within parenthesis in this document are US measurements.